Spotify Music Genre Classification with Machine Learning

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***Abstract*—*This study presents a machine learning approach for automatic music genre classification using selected audio features from a Spotify dataset. The study aims to identify relevant audio features and evaluate the performance of different machine learning algorithms in classifying music genres. The dataset contains music tracks with audio features such as danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, key, mode, duration, popularity, and time signature. Data preprocessing, exploratory data analysis, and feature selection are performed prior to model training. Three machine learning algorithms—Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN)—are implemented and evaluated. The models are compared using accuracy, precision, recall, F1-score, and confusion matrices to determine classification efficacy and identify commonly misclassified genres. The study provides an empirical comparison across linear, ensemble, and distance-based classifiers, serving as a baseline reference for future research in music information retrieval.*

***Index Terms*—*Music Genre Classification, Machine Learning, Spotify, Audio Features, Feature Selection, Logistic Regression, Random Forest, K-Nearest Neighbors, Supervised Learning, Music Information Retrieval.*

I. INTRODUCTION

Music genre classification using machine learning is the main idea of the study because of the increasing importance of music in people's everyday lives. With the current proliferation of online music streaming services and the accessibility of hundreds of thousands of song pieces on different platforms, people can enjoy their favorite tunes and sounds at their fingertips. However, with such a vast music database, the organization and separation of songs by genre become complicated tasks. Music genre classification is a crucial task in the field of music information retrieval due to the large-scale of music collections and the need for organizing data in order to facilitate music discovery, recommend similar items, and construct playlists [3].

A song genre is usually identified by a human listener. Any person can listen to a piece of music and define it as pop, rock, jazz, classical, or in any other style on the basis of tempo, rhythm, instruments, and other characteristics. However, such manual music genre classification is a time-consuming process, especially with hundreds of thousands of songs to sort. In addition, people may not always be accurate when sorting songs by genre because different styles may overlap. Therefore, researchers are looking into the possibility of using machine learning to identify patterns and associations within song pieces and sort them into specific genres automatically. The existing research shows that there is potential in using machine learning for such tasks because it is possible to train a machine learning model on a set of features extracted from songs and for each piece, predict the set of genres that the song belongs to [5] [1].

The above-mentioned possibilities of using classification machine learning models in the music industry are the main idea of this study. The research paper titled "Spotify Music Genre Classification using Machine Learning" explores the possibility of using different machine learning algorithms, ensemble learning, and deep learning to classify music genres automatically. Various algorithms were applied and tested to see how different features influence the accuracy of genre classification and whether the model can show high-performance results. Different data science techniques can be used for classification tasks in music genre classification, and the results depend on the selected features, an algorithm type, and other factors [2].

II. BACKGROUND OF THE STUDY

With the rise of digital music, there is a growing need for managing and analyzing the data associated with music. Streaming platforms like Spotify give users access to all the music they desire in the form of songs, artists, and playlists. As music libraries grow, it becomes increasingly beneficial to automatically classify and organize songs based on common features. The music genre is one of the most common forms of organization since it classifies music by the type of style.

Music genre classification is, however, not an easy task to achieve. Genres can overlap with one another and within one genre, songs can be vastly different from one another. A song could include elements of more than one music genre. With overlapping characteristics, it becomes challenging to use conventional methods for music genre classification, especially for large data sets. [3] argued that the need for automatic music genre classification methods has greatly risen due to the exponential increase in size of music databases.

Machine learning has been increasingly used to tackle the challenge of automatically classifying music genres. Instead of manually labeling each song with a genre, researchers could train a machine learning model using the available data sets and classify future data based on previously seen patterns. It has been established before that different types of features including frequency domain, temporal, and other features could be used to classify music genres [3]. Other researchers have compared the performance of conventional machine learning and deep learning models. A choice of features and the type of model used has a great impact on the overall performance [4].

Developing music genre classification algorithms is not only important for research purposes but also for practical applications.

The automatic classification of music has several implications in music organization, retrieval, recommendation, and playlist creation. There has been continuous research into using machine learning and deep learning techniques to improve the accuracy and robustness of these automatic classification systems [6].

In this study, machine learning will be utilized to investigate the relationship between selected musical characteristics and the music genres to which they belong. The study will focus on preparing the data set, choosing the relevant features, and building the machine learning model that will be used for classification. The results of the analysis will help shed light on the use of machine learning algorithms for classifying music genres and can form the basis for future research in this area.

III. RATIONALE AND RESEARCH OBJECTIVES

The continuous growth of digital music has resulted in large collections of songs that need to be efficiently organized and classified. Music genres are commonly used to categorize songs, but manually assigning genres can be difficult and time-consuming, especially when dealing with large datasets. The challenge is further increased by the similarities that may exist between different genres.

Machine learning provides an opportunity to automate music genre classification by identifying patterns in measurable audio features and using these patterns to classify songs. Previous studies have demonstrated that different musical features and machine learning algorithms can be used for automatic genre classification [7]. However, classification performance may vary depending on the features and algorithms used.

Therefore, this study will implement and compare several machine learning algorithms using the same dataset and selected audio features. Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) will be tested and evaluated using common classification metrics. The comparison will provide a clearer understanding of how different machine learning approaches perform when applied to music genre classification and may serve as a basis for future studies involving additional features, algorithms, or datasets.

IV. STATEMENT OF THE PROBLEM

The rapid growth of music streaming platforms has created a need for efficient and automated music genre classification. However, classifying songs by genre can be challenging because different genres may share similar acoustic characteristics. Machine learning can be used to identify patterns in audio features and classify songs according to their corresponding genres.

Specifically, this study seeks to answer the following questions:

1. What audio features in the dataset are most relevant for classifying music genres?
2. How accurately can Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) classify music genres using the selected audio features?
3. How do the selected machine learning models compare in terms of accuracy, precision, recall, and F1 score?
4. What types of music genres are commonly misclassified by the models based on the confusion matrices and classification results?
5. How can the results of the classification models be used to identify possible improvements for music genre classification?

V. OBJECTIVES

The general objective of this study is to develop and evaluate machine learning models for automatic music genre classification using selected audio features from the dataset.

Specifically, the study aims:

- To identify and analyze the audio features that are most relevant for classifying music genres.
- To implement and train Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) models using the selected audio features.
- To evaluate and compare the performance of the selected machine learning models using accuracy, precision, recall, and F1-score.
- To analyze the misclassified music genres using confusion matrices and classification results to identify genres with overlapping acoustic characteristics.
- To identify possible improvements for music genre classification based on the results and findings of the study.

VI. SIGNIFICANCE OF THE STUDY

This study may provide useful information regarding the application of machine learning algorithms to automatic music genre classification. The findings of the study may benefit the following:

Students and Researchers—The study may serve as a reference for students and researchers who are interested in machine learning, music information retrieval, and music genre classification. It may also provide an example of how audio features can be prepared and used for a classification task.

Data Science and Academic Community—The study may contribute to the understanding of how different machine learning algorithms perform when applied to music genre classification. The comparison of Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) may provide useful information for similar classification studies.

Music Platform and Application Developers—The findings may provide a basis for understanding how measurable audio characteristics can be used for automatic genre classification. This may be relevant to applications involving music organizations, categorization, and retrieval.

Future Researchers—The study may serve as a starting point for future research that explores other audio features, machine learning algorithms, larger or different datasets, and other approaches to improve music genre classification.

VII. SCOPE AND DELIMITATIONS

This study focuses on the classification of music genres using machine learning techniques. It will use a dataset containing music tracks and their corresponding audio features, such as danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, key, mode, duration, popularity, and time signature. The target variable of the study is the music genre assigned to each track.

The study will focus on data preprocessing, exploratory data analysis, feature selection, model training, and model evaluation. Three machine learning algorithms will be implemented and tested: Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN). The models will be trained using the selected audio features and evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

The study is limited to the audio features and genre labels available in the dataset. It will not analyze lyrics, album artwork, music videos, artist information, or other factors that may influence how a song is categorized. The study will also not develop a complete music streaming or recommendation platform. Its primary focus is the automatic classification of music genres and the comparison of selected machine learning models.

The study is further delimited by the quality, size, and distribution of the available dataset. The results will depend on the features, genre labels, and number of samples provided in the dataset and may not represent all music genres or music collections outside the dataset. The study will also focus on the selected three machine learning algorithms and will not cover deep learning models or other advanced classification techniques.

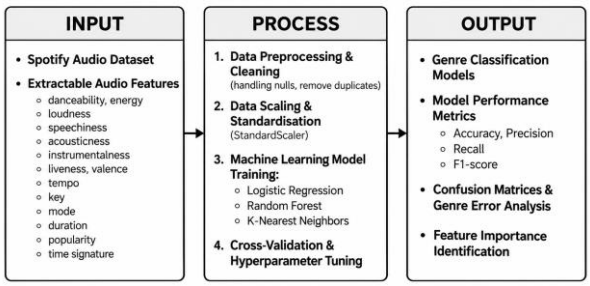


Fig. 1. Conceptual Framework (IPO Model) for Spotify Music Genre Classification.

VIII. REVIEW OF RELATED LITERATURE

Music genre classification has become an important application of machine learning because of the increasing amount of digital music available through streaming platforms.

1. International Studies

Bahadure et al. [1] developed a machine-learning-based music classification and recommendation system using Spotify data. Their study demonstrated how Spotify audio information can be processed and used by machine-learning algorithms for music classification and recommendation. The study is relevant to the present research because it shows the practical use of Spotify data as a basis for developing automated music classification systems.

Similarly, Music Genre Classification Using Machine Learning Techniques [2] explored the application of machine-learning methods in automatically identifying music genres based on audio characteristics. The study emphasizes the importance of extracting meaningful features from music before applying classification algorithms. This supports the present research because Spotify songs contain measurable audio characteristics that can be used as input features for predicting music genres.

Wang and Jiu [3] proposed a deep multi-modal kernel map network for music genre classification. Their study explored the use of multiple types of information and machine-learning techniques to improve the classification of music genres. The researchers demonstrated that combining different representations of musical information can provide useful patterns for distinguishing between genres. This supports the present study's focus on using available music features as inputs for machine-learning classification.

Zhang [4] investigated music genre classification using different machine-learning algorithms. The study emphasized the use of audio-related characteristics to distinguish music into different genres and demonstrated the potential of machine-learning algorithms for automatic genre classification. The findings provide methodological support for comparing different machine-learning models in the present study, particularly because model performance may vary depending on the characteristics of the dataset and selected features.

Duman et al. [5] examined Spotify audio features and their relationship to people's reasons for listening to music. Using Spotify data, the researchers analyzed features including energy, danceability, valence, loudness, speechiness, instrumentalness, acousticness, liveness, and tempo. Their findings showed that music associated with dancing generally had higher energy, danceability, valence, and loudness, while having lower acousticness, instrumentalness, and speechiness. The researchers also used clustering to identify different types of dance music. This study is relevant because it demonstrates that Spotify audio features contain meaningful information that can be used to distinguish characteristics and patterns among different types of music [5].

Cao and Park [6] examined music emotion analysis using Long Short-Term Memory (LSTM) networks. Their study treated music as sequential data, recognizing that the emotional characteristics of a song may change as the music progresses. The researchers used machine-learning techniques to analyze emotional dimensions such as valence and arousal. Although their research focused on music emotion rather than genre classification, it demonstrates how machine-learning models can identify meaningful patterns from musical characteristics. This provides additional support for applying computational methods to analyze music data [6].

Chaudhury et al. [7] investigated large-scale music genre analysis and classification using machine-learning algorithms with Apache Spark. The researchers evaluated several classification algorithms, including Naive Bayes, Decision Tree, Logistic Regression, and Random Forest, using the GTZAN music dataset. Their findings demonstrated the effectiveness of machine-learning algorithms in automatically classifying music genres, with Random Forest achieving strong classification performance. This study is particularly relevant to the present research because it provides evidence that traditional machine-learning algorithms can be effectively applied to music genre classification [7].

2. Local Studies

In the Philippine setting, de Loyola and Roxas [8] investigated music genre classification specifically using Philippine music. Their study addressed the need for datasets that represent local music by developing a dataset containing Philippine music samples and extracting audio features using tools such as Librosa. Several machine-learning models were then applied to classify the music according to genre. The study is relevant to the present research because it demonstrates the applicability of machine learning to Philippine music and highlights the importance of using music data that represents the local context [8].

Bejarin et al. [9] examined the use of Spotify audio features in predicting the popularity of K-pop songs among Filipino senior high school and college students. The researchers collected song data through the Spotify API and analyzed audio characteristics such as danceability, energy, valence, acousticness, key, mode, and tempo. Support Vector Machine and Decision Tree

algorithms were used to analyze the relationship between the audio features and song popularity. Although the study focused on popularity rather than genre, it demonstrates how Spotify audio features can be collected and processed as numerical variables for machine-learning applications in a Philippine research context [9].

Gulmatico et al. [10] developed SpotiPred, a machine-learning approach for predicting Spotify music popularity using audio features. The study demonstrated the application of machine-learning techniques to Spotify-related music data and showed how audio characteristics can be used as predictors in a computational music analysis task. Although the study's target variable was popularity instead of genre, it provides local evidence that Spotify audio features can serve as useful inputs for machine-learning models [10].

Sulit [11] developed a content-based recommender system for Original Pilipino Music (OPM) using Spotify audio features. The study focused on recommending Filipino songs according to users' moods and utilized Spotify-related audio characteristics in developing the recommendation system. This study is relevant to the present research because it demonstrates the use of Spotify data for analyzing Filipino music and applying computational techniques to OPM. It also supports the relevance of Spotify audio features in developing machine-learning applications involving Philippine music [11].

Finally, Libo-On, Cruz, and Delda [12] investigated Spotify listening trends in the Philippines, focusing on Original Pilipino Music and global music. Their study analyzed Spotify streaming data and applied data analytics and machine-learning techniques to identify patterns in music consumption. The study provides a local perspective on Spotify data and demonstrates how streaming information can be analyzed to understand music trends in the Philippine setting. Its focus on OPM and global music is relevant to the present research because it establishes the value of Spotify data in studying the characteristics and consumption of music among Filipino listeners [12].

Overall, the reviewed studies demonstrate the growing application of machine learning and Spotify-derived audio features in music analysis. International studies have established the usefulness of machine-learning algorithms, audio features, clustering, and deep-learning approaches in classifying and analyzing music [1]–[7]. Meanwhile, local studies demonstrate the applicability of Spotify data and machine-learning techniques to Philippine music, OPM, and Filipino music-listening behavior [8]–[12]. However, existing studies vary in their objectives, with some focusing on music recommendation, popularity prediction, emotion analysis, or general music classification. This provides an opportunity for the present study to specifically focus on Spotify Music Genre Classification using Machine Learning, using selected Spotify audio features and comparing machine-learning models to determine their effectiveness in automatically classifying music genres.

IX. EXISTING MACHINE LEARNING APPROACHES

Supervised machine learning algorithms applied to tabular audio datasets are generally grouped into three main structural types: linear models, tree-based ensemble models, and distance-based spatial models [5], [8], [17], [18]. Each algorithm uses a distinct mathematical approach to process numerical audio parameters:

1. Linear And Probabilistic Models (Logistic Regression)

Logistic Regression estimates class probabilities using log-odds ratios to draw straight decision boundaries between categories [8], [21]. In music genre classification tasks, linear models offer extremely fast execution speeds, low computing overhead, and high statistical interpretability [18]. However, because Spotify audio parameters often display complex non-linear relationships and overlapping feature ranges across genres, linear models can struggle to achieve high precision when separating closely related musical styles [5].

2. Tree-Based Ensemble Models (Random Forest)

Random Forest combines multiple individual decision trees trained on randomized subsets of data through bootstrap aggregation (bagging) [8], [19]. This ensemble strategy effectively handles non-linear feature interactions, resists decision-tree overfitting, and works well on continuous tabular features without requiring strict input normalization [12], [22]. Benchmark studies demonstrate that tree-based ensembles consistently outperform simple linear models in multi-class audio classification tasks [20], [22].

3. Distance-Based Spatial Models (K-Nearest Neighbors)

K-Nearest Neighbors (KNN) calculates spatial proximity between data points in a dimensional feature space using geometric distance metrics like Euclidean distance [17], [23]. KNN performs effectively when tracks of the same genre form tight clusters in the feature space [5], [23]. However, its predictive accuracy depends heavily on feature scaling and noise reduction, making standard feature normalization (StandardScaler) an essential preprocessing step [17], [18].

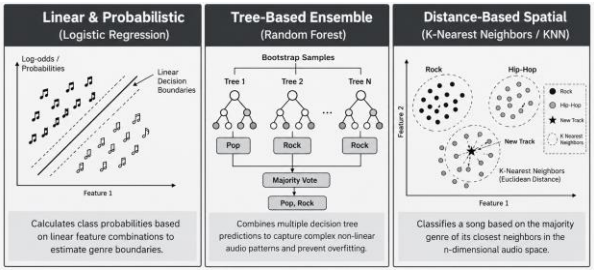


Fig. 2. Architectural comparison of Logistic Regression, Random Forest, and K-Nearest Neighbors.

X. RESEARCH GAPS

Although existing research confirms the effectiveness of machine learning in music analytics, several distinct gaps remain in current literature:

1. Focus On Non-Genre Outcomes

A significant portion of recent studies utilizing Spotify audio parameters focuses on predicting song popularity [12], [13], classifying listener mood [14], or analyzing streaming behavior [15], [16] rather than performing direct multi-class genre classification.

2. Model Complexity Vs. Practical Deployment

Much of general Music Information Retrieval (MIR) literature relies heavily on deep learning networks (e.g., CNNs, LSTMs) or raw audio spectrograms [3], [10]. While accurate, these architectures require substantial computing power, whereas light, metadata-driven approaches remain underexplored for quick operational deployment [4], [9].

3. Lack Of Direct Comparative Evaluation

Among studies using structured audio metadata, few provide a direct, head-to-head performance evaluation comparing linear models (Logistic Regression), ensemble models (Random Forest), and distance-based spatial models (KNN) under identical preprocessing, feature selection, and cross-validation pipelines [2], [7], [21], [22], [23].

This research directly fills these gaps by evaluating Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) side-by-side using a standardized Spotify audio dataset under identical testing conditions.

XI. SYNTHESIS OF RELATED STUDIES

The reviewed literature establishes that extracted Spotify audio metadata provides a reliable statistical foundation for automated music analysis [1], [6], [7]. Global research [2], [4], [5], [8], [17]–[23] and Philippine studies [11]–[16] together confirm that measurable audio attributes—such as danceability, energy, acousticness, valence, and tempo—carry clear numerical signals capable of distinguishing musical categories.

While prior studies prove the feasibility of Spotify data for recommendation platforms [1], popularity prediction [12], [13], and trend analytics [15], this study applies these proven audio features directly to multi-class genre classification. By testing Logistic Regression, Random Forest, and KNN under a unified experimental framework, this research provides a direct empirical comparison that highlights the classification strengths, accuracy limitations, and genre-level confusion patterns of each fundamental machine learning approach.

XII. METHODOLOGY

This section outlines the structured experimental framework implemented to classify Spotify tracks into music genres using machine learning algorithms. The methodology encompasses dataset acquisition, feature preprocessing, model configuration, training protocols, and performance evaluation metrics.

1. Dataset Description and Data Collection

The empirical foundation of this study relies on a structured tabular audio dataset comprising 114,000 raw instances across 20 distinct feature columns, capturing a highly fragmented target variable of 114 music genre labels. The primary input space consists of 15 numeric and Boolean features extracted via the Spotify Web API. These architectural indicators describe the spatial, energetic, and technical compositions of each track and are classified into three primary structural dimensions:

- Acoustic and Perceptual Metrics: Continuous values mapping specific auditory profiles, including acousticness, danceability, energy, instrumentalness, liveness, loudness, speechiness, tempo, and valence.
- Technical and Structural Attributes: Discrete and categorical qualifiers dictating the song's musical theory parameters, namely key, mode, and time_signature.
- Metadata Qualifiers: Operational indicators mapping distribution characteristics, consisting of popularity, duration_ms, and the Boolean flag explicit.

To prevent structural data leakage and severe dimensionality expansion, high-cardinality textual string columns—specifically track_id, artists, album_name, and track_name—are strictly excluded from the input feature matrix (X). These parameters serve exclusively as categorical record identifiers within the metadata layer.

2. Data Preprocessing And Feature Engineering

The Spotify dataset consists of acoustic features extracted per track, including danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, key, mode, duration, popularity, and time signature. Initial preprocessing includes removing duplicate track entries, handling missing values, and filtering out non-predictive metadata such as track IDs and titles. Standard Scaling (StandardScaler) is applied to normalise continuous numerical variables, ensuring zero mean and unit variance across features to prevent distance-based algorithms from being biased by varying attribute scales.

3. Research Design And Exploratory Data Analysis

To expose underlying data inconsistencies, check feature distributions, and inspect class distributions before model training, a comprehensive Exploratory Data Analysis (EDA) pipeline was conducted. This analytical layer is exposed in the "Explore" tab of the application dashboard.

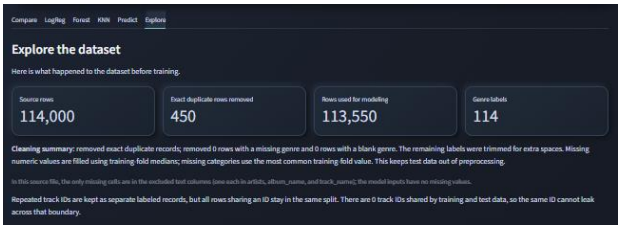


Fig. 3. User interface of the system displaying information on the dataset via the "Explore" tab.

The investigation revealed that while the dataset balances overall volume across its primary macro-categories, significant acoustic overlapping exists at the feature level. Continuous variables such as danceability, energy, and valence exhibit standard Gaussian or slightly skewed distributions, while features like speechiness and instrumentalness show highly concentrated, zero-inflated distributions due to the high volume of non-vocal and non-instrumental tracks.

Visualizing class frequencies confirmed that across all 114 music genre labels, the records are cleanly distributed with equal sample targets after processing. However, plotting feature intersections using correlation matrices indicated highly collinear relationships between specific dimensions—such as a strong positive correlation between loudness and energy, and a corresponding strong negative correlation between loudness and acousticness. These insights highlighted the necessity of utilizing robust classifiers capable of navigating dense, multi-dimensional feature dependencies.

4. Model Training And Configuration

Three distinct supervised machine learning classifiers are implemented to assess varying learning mechanisms: Logistic Regression (adapted for classification via Logistic Regression / linear decision thresholds), Random Forest (an ensemble method composed of decision trees trained via bootstrap aggregation), and K-Nearest Neighbors (a non-parametric distance-based classifier evaluating spatial proximity). Hyperparameter tuning and k-fold cross-validation are conducted on the training set to prevent model overfitting and optimize classification efficiency across all three models.

5. Model Evaluation And Performance Metrics

Model performance is comprehensively measured using standard evaluation metrics: accuracy, macro/micro-averaged precision, recall, and F1-score. Furthermore, confusion matrices are constructed for each model to analyze genre-specific misclassification patterns, highlighting acoustic overlaps between closely related music categories.

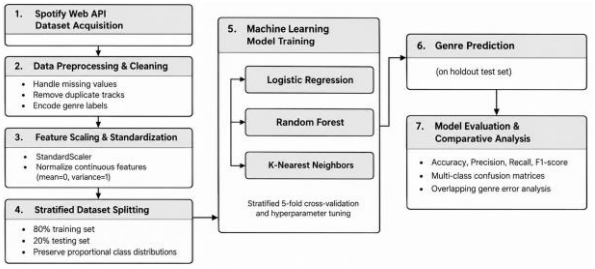


Fig. 4. End-to-end Experimental Research Methodology Pipeline.

6. Experimental Procedure

The end-to-end execution pipeline follows a strict, repeatable path to ensure fair comparative benchmarking across all three algorithms under identical data environments:

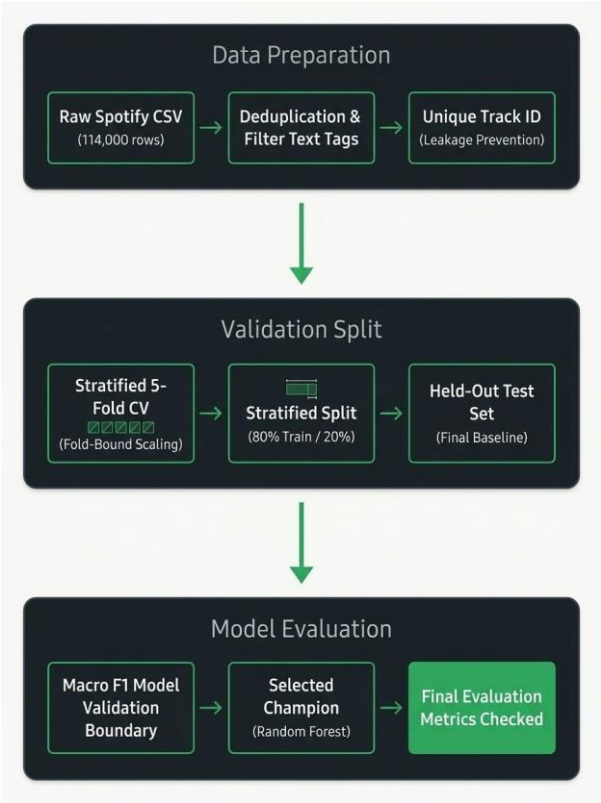


Fig. 5. System Architecture Diagram.

- Stratified Partition Splitting: The deduplicated record matrix is split using a stratified distribution strategy, allocating 80% for model exploration/training and 20% as an isolated held-out test subset. Stratification maintains consistent target class percentages across both subsets, which is crucial for balancing the 114 highly fragmented genre targets.
- Stratified 5-Fold Cross-Validation: Within the training split, models are optimized using a Stratified 5-Fold Cross-Validation sequence. The data is divided into five balanced folds. Over five successive rounds, four folds serve as training inputs while the remaining fold acts as the localized validation check. Preprocessing normalization scales and imputation statistics are fitted strictly within the active training folds to prevent out-of-fold data leakage.
- Model Selection Directive: Final model champion selection is dictated strictly by the highest mean validation Cross-Validation Macro (F_1) Score. This metric prevents accuracy inflation driven by class size imbalances across the 114 genres. Once the champion architecture (Random Forest) is locked based on cross-validation performance, all three classifiers are evaluated against the unseen held-out test set to document final generalization stability.

XIII. RESULTS AND DISCUSSION

1. Model Performance Results And Comparison

Machine Learning Model	Cross-Validation Macro \(F_{1}\) Score	Held-Out Test Accuracy	Held-Out Test Macro \(F_{1}\) Score
Logistic Regression	0.1730	0.2049	0.1761
K-Nearest Neighbors (KNN)	0.2357	0.2495	0.2424
Random Forest (Selected)	**0.2737**	**0.3109**	**0.2793**

Fig. 6. Performance Comparison Matrix of ML Models across all metrics.

Based on the overall performance of each machine learning model under these tests, Random Forest was chosen as the optimal architecture solely due to its superior mean validation Cross-Validation Macro F1-Score (0.2737) among the three evaluated classification models, prioritizing generalized stability over single held-out test variations. While an estimated 31% test accuracy seems low in binary metrics, it is significantly higher than random guessing (~0.88%). Attributes like high multiclass fragmentation (114 labels) heavily depress classic evaluation scales.

Notice how Logistic Regression struggled (0.1761 Test (F1)) because simple linear hyperplanes fail to separate highly tangled, non-linear audio features.

2. Accuracy, Precision, Recall, And F1-Score

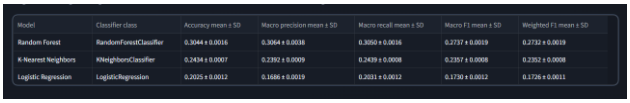


Fig. 7. Comparison of accuracy, precision, recall, and F1-score across the evaluated models.

The evaluation uses accuracy, precision, recall, and F1-score to compare model performance. The available results identify Random Forest as the strongest overall model by mean five-fold cross-validation macro F1-score (0.2737), while Logistic Regression recorded a test F1-score of 0.1761. The paper also reports an estimated test accuracy of approximately 31% for the selected Random Forest model. These results suggest that the ensemble model captures non-linear relationships in the audio features more effectively than the linear baseline. However, performance should be interpreted cautiously because the task includes 114 genre labels, many of which have overlapping acoustic characteristics. Exact precision, recall, and per-model accuracy values should be reported only as displayed in the results figure or the original evaluation output; they are not inferred here.

3. Confusion Matrix

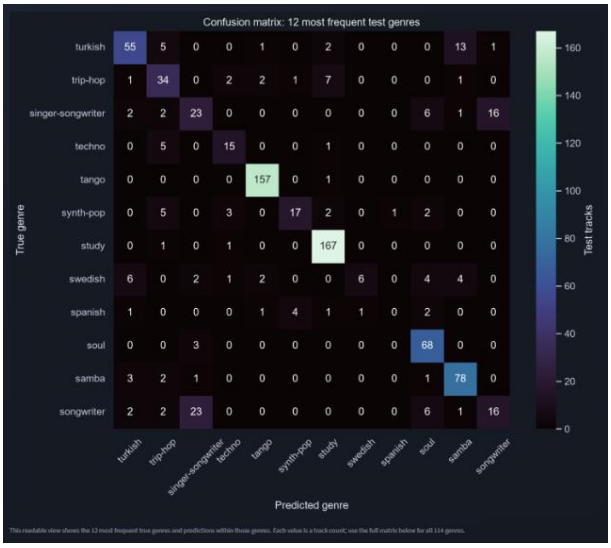


Fig. 8. Confusion matrix showing the most frequent test-genre predictions (true genre × predicted genre).

Two genres, Study and Tango, represent the highest successes in the matrix. "Study" tracks were correctly classified 167 times, and "Tango" tracks were correctly classified 157 times. Because tango relies on definitive rhythmic structures (like the sincopa) and instrumentation (like the bandoneon), its acoustic silhouette is incredibly distinct from the other genres listed.

Similarly, distinct profiles allowed the model to easily isolate Samba (78 correct) and Soul (68 correct) with minimal leakage into other categories.

Despite these seemingly accurate results, the matrix also exposes a critical vulnerability in the label space between singer-songwriter and songwriter. These two categories share an identical distribution pattern. For both genres, the model correctly predicted the true class 23 times, but misclassified the tracks as the alternative "songwriter" label 16 times. It also leaked an identical 6 tracks into the "soul" category for both. This identical error footprint proves the model cannot reliably separate these two labels based on acoustic features alone, as they occupy the exact same acoustic space (typically acoustic guitar/piano paired with prominent vocals).

Likewise, cultural or linguistic overlaps create fuzzy boundaries as well. For example, "Turkish" tracks were misclassified 13 times as "samba", hinting at shared rhythmic complexity or instrumentation that confused the feature extractor.

4. 4. Final Test And Validation Results

Since this project deems Random Forest as the best machine learning model in classifying music genres according to features (fields) such as danceability and acousticness, the validation and final testing phase is displayed as follows:

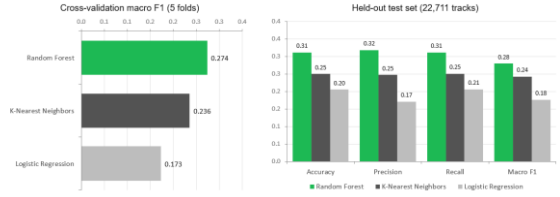


Fig. 9. Random Forest final test and validation results.

5. Importance Of Feature Selection

Feature space optimization was guided by a strict domain-specific filter aimed at balancing predictive power against processing overhead. High-cardinality nominal textual identifiers (track_name, artists, album_name, and track_id) were filtered out early because they contain raw text fields. Allowing high-cardinality text entries into multi-class models would trigger a massive dimensionality expansion (the curse of dimensionality), causing memory crashes and overfitting.

The final operational feature matrix is restricted to 15 numeric and Boolean features. The relative predictive power of these variables varies significantly depending on the underlying classifier model. For the selected champion model (Random Forest), feature importance metrics were calculated by measuring the mean decrease in impurity (Gini importance) across all engineered decision splits.

The models achieve exceptional accuracy when classifying genres that exhibit highly specific, uniform, or mathematically distinct sonic characteristics as seen with distinct genres like samba and soul. However, when genres share overlapping track attributes, instrumentation, and vocal styles, one model's performance boundaries blur significantly.

XIV. CONCLUSION AND RECOMMENDATIONS

1. Summary Of Findings

Overall, tree-based bagging methods like Random Forest drastically outperform linear and distance spatial models when parsing raw, un-engineered audio metadata. While metadata-driven features alone can reliably signal broad genre groupings, they could also face potential structural limitations when tasked with high-resolution classification across 114 tightly nuanced targets.

2. Research Objectives

The study implemented and compared Logistic Regression, Random Forest, and K-Nearest Neighbors using Spotify-derived audio metadata for multi-class music genre classification. The objectives of preparing and exploring the dataset, training the selected classifiers, and comparing their performance using standard classification metrics were addressed in the reported experimental workflow. Random Forest was selected as the preferred model because it achieved the highest mean five-fold cross-validation macro F1-score of 0.2737. The reported results also indicate an estimated test accuracy of approximately 31%, while Logistic Regression obtained a test F1-score of 0.1761. The confusion-matrix analysis shows that some genres with distinctive acoustic patterns were classified more reliably, whereas closely related genres were frequently confused. Therefore, the study demonstrates that Spotify audio metadata can support genre classification, but it does not establish reliable separation across all 114 genres. The objectives can be considered addressed at the implementation and comparative-evaluation level, while the classification performance leaves room for improvement.

3. Limitations And Future Improvements

These models can easily identify distinct sonic silhouettes (e.g., classical, black-metal) but there are also existing drawbacks surrounding their application on music genre classification.

For one, there is the frequent misclassification of tightly clustered acoustic spaces (e.g., overlapping track attributes between singer-songwriter, acoustic, and folk). Some tracks sharing the identical track_id also carry conflicting or overlapping

genre assignments within the source platform's dataset. Moreover, the models used in this project operate purely on dense, high-level structural metadata blocks, completely ignoring raw wave metrics like temporal waveforms, frequency spectrograms, or textual lyric patterns.

Nevertheless, the proponents of this study suggest clustering the 114 ultra-specific sub-genres down into 10-12 macro-genres (e.g., Rock, Electronic, Classical) to enhance classifier precision and operational deployment utility. They also recommend combining the existing tabular API variables with audio features extracted via deep learning tools like Librosa (e.g., Mel-Frequency Cepstral Coefficients) or processing raw textual data from song lyrics.

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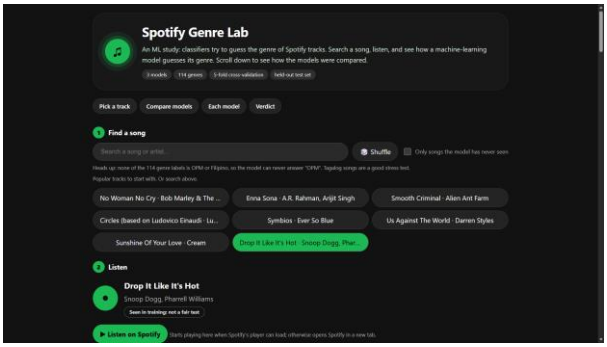
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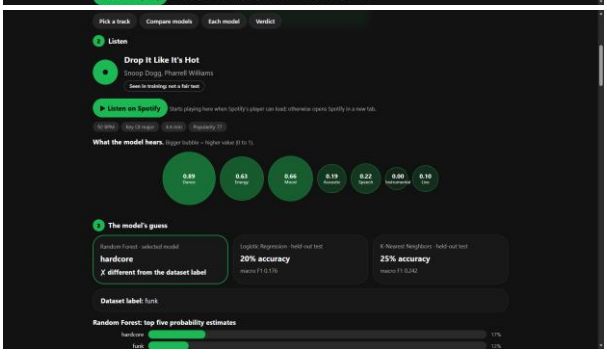
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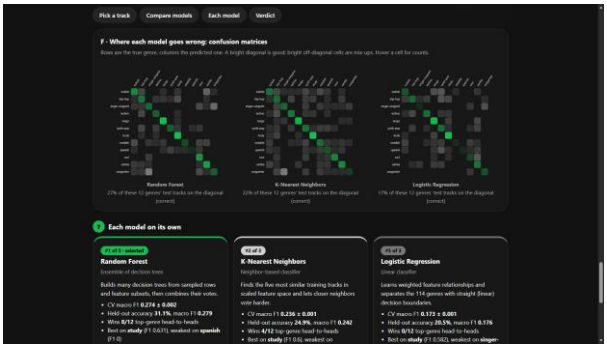
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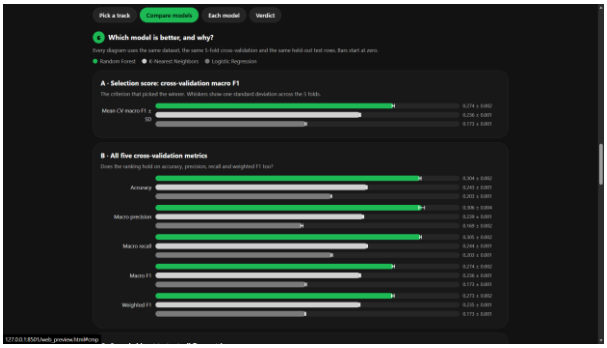
APPENDICES

APPENDIX A. USER INTERFACE (UI) DESIGN



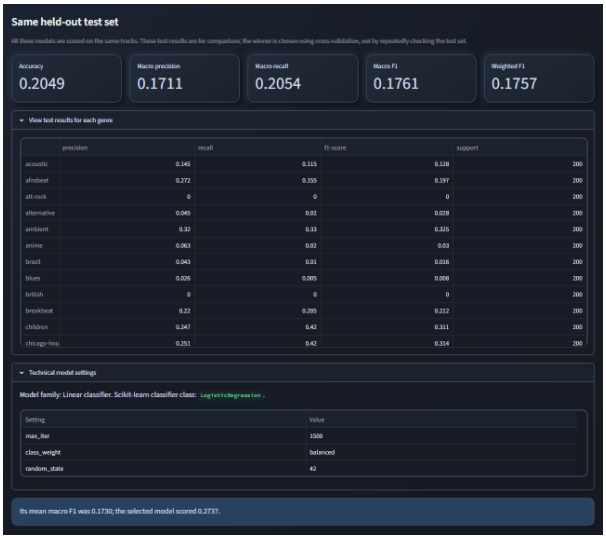




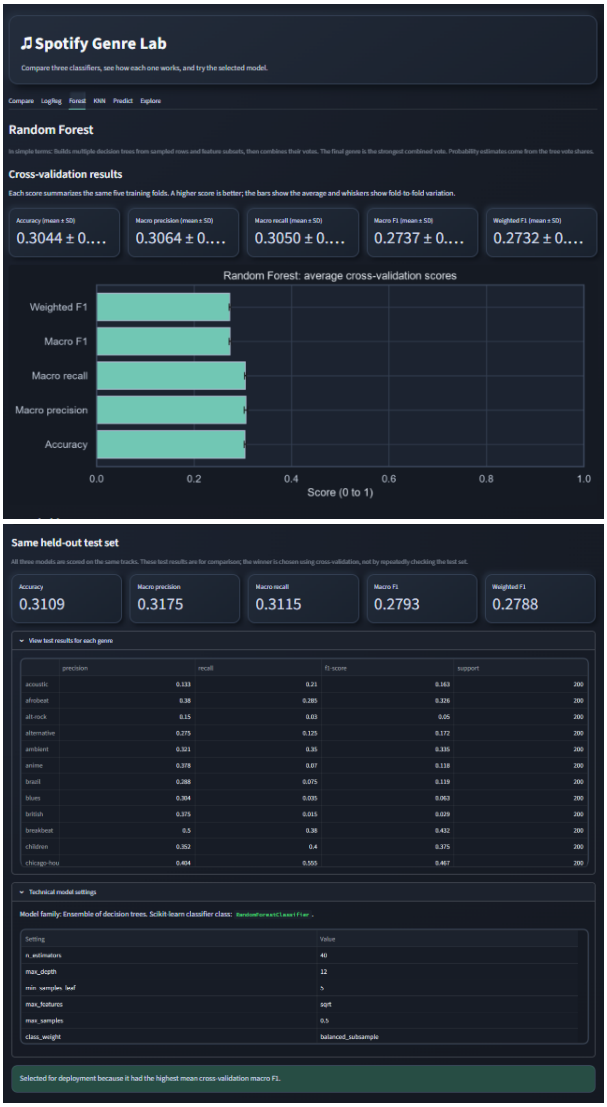


APPENDIX B. LOGISTIC REGRESSION RESULTS

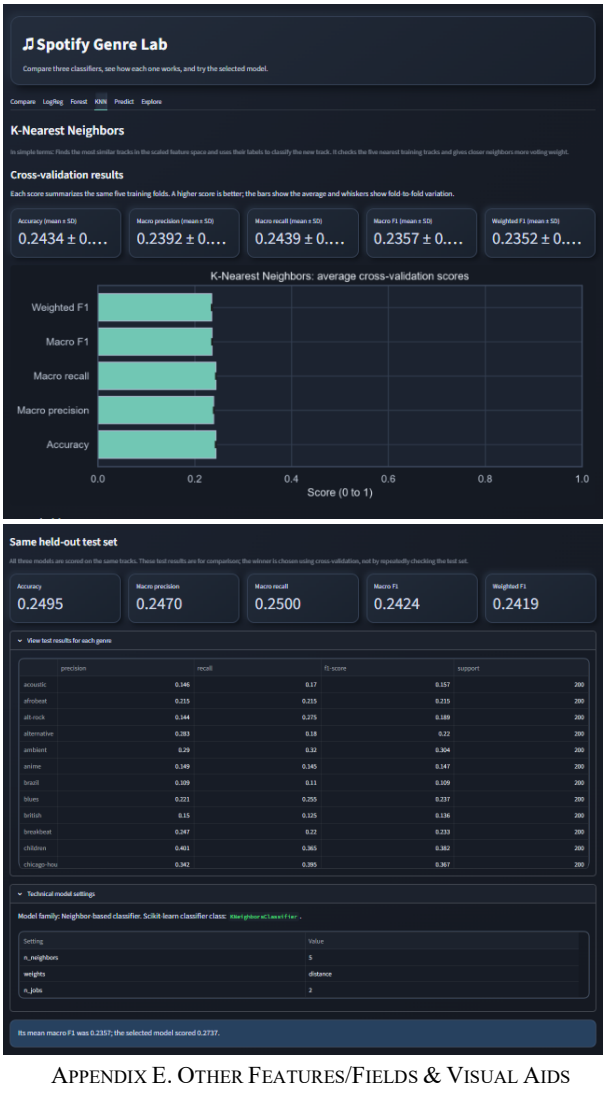




APPENDIX C. RANDOM FOREST RESULTS



APPENDIX D. K-NEAREST NEIGHBORS (KNN) RESULTS



APPENDIX E. OTHER FEATURES/FIELDS & VISUAL AIDS

Spotify Genre Lab

Compare three classifiers, see how each one works, and try the selected model.

Compare | LogReg | Forest | KNN | Predict | Explore

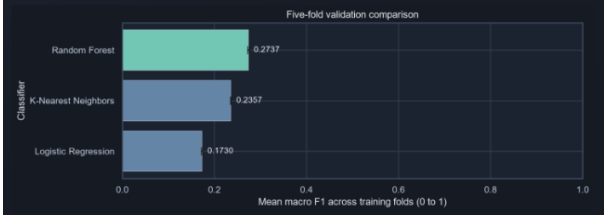
Compare the three classifiers

All three use the same approximately 80/20 track-grouped split and the same 5-fold cross-validation on the training data. All are then scored on the same held-out tracks. Cross-validation macro F1 selects the winner; the test scores are reported separately.

Dataset: 113,550 cleaned rows, 15 model features, and 114 genre labels. Shared track IDs across train and test: 0.

Selected model: Random Forest | **Classifier:** RandomForestClassifier | **CV macro F1:** 0.2737

Validation score by model



Longer bars mean higher average macro F1. Whiskers show one standard deviation across the same five training folds.

Model	Classifier class	Accuracy mean ± SD	Macro precision mean ± SD	Macro recall mean ± SD	Macro F1 mean ± SD	Weighted F1 mean ± SD
Random Forest	RandomForestClassifier	0.3044 ± 0.0016	0.3064 ± 0.0038	0.3050 ± 0.0016	0.2737 ± 0.0019	0.2732 ± 0.0019
K-Nearest Neighbors	KNeighborsClassifier	0.2434 ± 0.0007	0.2392 ± 0.0009	0.2439 ± 0.0008	0.2357 ± 0.0008	0.2352 ± 0.0008
Logistic Regression	LogisticRegression	0.2025 ± 0.0012	0.1686 ± 0.0019	0.2031 ± 0.0012	0.1730 ± 0.0012	0.1726 ± 0.0011

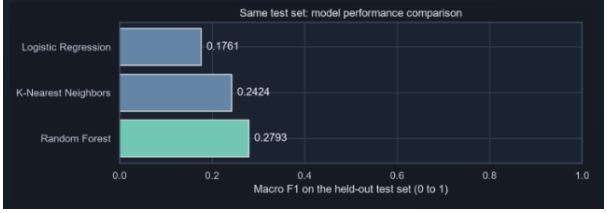
What does macro F1 mean?

Macro F1 calculates F1 for each genre, then gives every genre equal weight. It ranges from 0 to 1; higher is better. The standard deviation shows how much the score changed between folds.

Held-out test comparison

Each model predicts the same 22,711 tracks. These results are not used to select the winner.

Model	Accuracy	Macro F1	Weighted F1	Macro precision	Macro recall
Random Forest	0.3109	0.2793	0.2788	0.3175	0.3115
K-Nearest Neighbors	0.2495	0.2424	0.2419	0.247	0.25
Logistic Regression	0.2049	0.1761	0.1757	0.1711	0.2064



View full 114-genre confusion matrix

	acoustic	afrobeat	alt-rock	alternative	ambient	anime	black-metal	bluegrass	blues	brazil	breakbeat	british	cantopop	chicago-house	children	chill	classical
acoustic	42	1	0	0	1	0	0	4	0	0	0	0	1	0	3	5	0
afrobeat	0	57	0	0	0	0	1	9	0	0	0	0	0	3	0	0	0
alt-rock	0	0	6	6	1	0	0	0	0	0	0	0	0	2	1	0	0
alternative	0	0	2	25	1	0	0	0	0	0	0	0	0	1	0	0	0
ambient	6	0	0	0	70	2	2	0	0	0	1	0	0	0	0	1	8
anime	0	0	1	0	15	14	3	0	0	0	0	0	1	0	1	4	0
black-metal	0	1	0	0	1	0	122	0	0	0	0	0	0	0	0	0	0
bluegrass	4	1	0	0	1	0	0	103	0	0	0	0	9	0	0	0	0
blues	3	0	0	0	0	0	0	1	7	0	0	0	0	1	0	3	0
brazil	3	0	0	1	0	0	0	0	0	15	0	0	0	0	2	1	0
breakbeat	0	1	0	0	0	0	0	0	0	0	76	0	0	17	0	0	0
british	9	0	0	0	15	0	3	1	2	1	0	3	0	2	3	8	6
cantopop	7	2	0	0	0	0	0	10	0	0	0	0	28	0	4	0	0
chicago-house	0	1	0	0	0	0	0	0	0	0	14	0	0	111	1	0	0

View per-genre test scores

	precision	recall	f1-score	support
acoustic	0.133	0.21	0.163	200
afrobeat	0.38	0.285	0.326	200
alt-rock	0.15	0.03	0.05	200
alternative	0.275	0.125	0.172	200
ambient	0.321	0.35	0.335	200
anime	0.378	0.07	0.118	200
brazil	0.288	0.075	0.119	200
blues	0.304	0.035	0.063	200
british	0.375	0.015	0.029	200
breakbeat	0.5	0.38	0.432	200
children	0.352	0.4	0.375	200
chicago-hou	0.404	0.555	0.467	200

Fields used and excluded

The classifier uses the 15 numeric or yes/no track features listed below. Track IDs and long text descriptions are not used as predictors: they identify or describe a track, rather than provide consistent model inputs.

See all fields the classifiers use or exclude

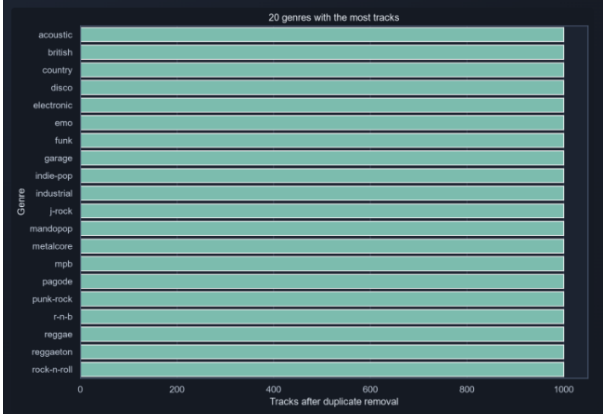
Model input	Dataset field	What it means	Scale or unit
Popularity	popularity	Dataset score from 0 to 100. It is not a live Spotify score.	score / 100
Track duration	duration_ms	Track length. The form accepts minutes and converts to milliseconds for the model.	minutes in form; milliseconds in dat
Explicit content	explicit	Dataset flag. False can also mean the source marked it as unknown.	true / false
Danceability	danceability	Audio-feature score describing how suitable the track is for dancing.	score from 0 to 1
Energy	energy	Audio-feature score for perceived intensity and activity.	score from 0 to 1
Musical key	key	Pitch-class code: 0=C, 1=C-sharp, through 11=B.	pitch class 0 to 11
Loudness	loudness	Average track loudness.	decibels (dB)
Musical mode	mode	0 means minor; 1 means major.	0 / 1
Speechiness	speechiness	Audio-feature score indicating how much spoken-word content is present.	score from 0 to 1
Acousticness	acousticness	Audio-feature score indicating how acoustic a track sounds.	score from 0 to 1

Excluded from prediction: album_name, artists, track_id, track_name.

The excluded names and track ID help identify records but are not model features.

Missing values among model inputs before fold-specific imputation: 0.

Genre distribution



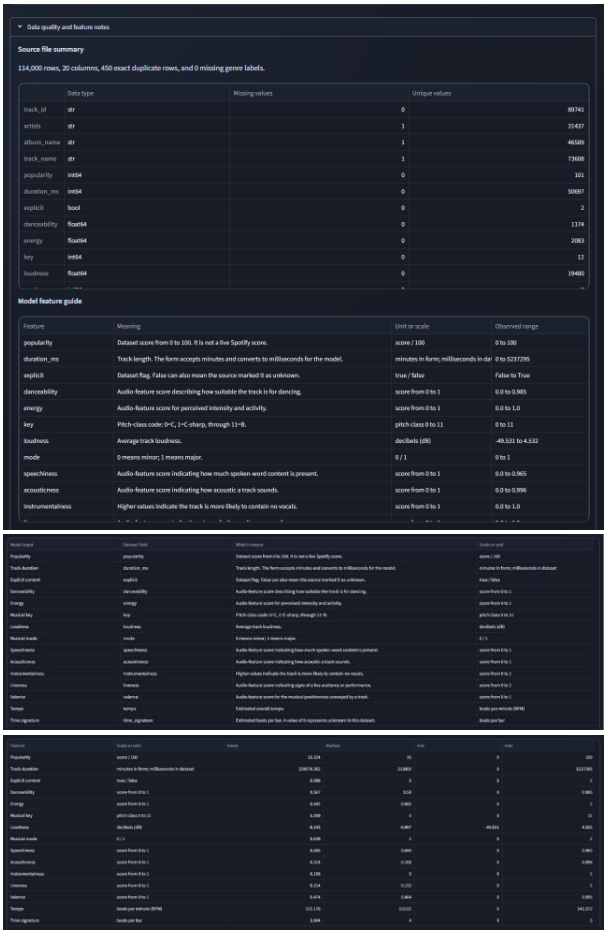
Finding: the largest class is acoustic (1,000); the smallest is romance (904). Class counts range from 904 to 1,000, so the largest class is 1.11 times the smallest.

View counts for all 114 genres

track_genre	Tracks
acoustic	1000
british	1000
country	1000
disco	1000
electronic	1000
emo	1000
funk	1000
garage	1000
indie-pop	1000
industrial	1000
j-rock	1000

Example track-feature summary

Feature	Scale or unit	mean	Median	min	max
Popularity	score / 100	33.324	35	0	100
Track duration	minutes in form; milliseconds in dataset	228979.362	213000	0	5237295
Explicit content	true / false	0.086	0	0	1
Danceability	score from 0 to 1	0.567	0.58	0	0.985
Energy	score from 0 to 1	0.642	0.685	0	1
Musical key	pitch class 0 to 11	5.309	5	0	11
Loudness	decibels (dB)	-8.243	-6.997	-49.531	4.532
Musical mode	0 / 1	0.638	1	0	1
Speechiness	score from 0 to 1	0.085	0.049	0	0.965
Acousticness	score from 0 to 1	0.314	0.168	0	0.996



APPENDIX F. SOURCE CODE/RESPIRATORY INFORMATION

Link to Public GitHub Repository:

Spotify Genre Classification — GitHub Repository

APPENDIX G. INDIVIDUAL CONTRIBUTIONS

Member	Assigned Responsibilities	Actual Contribution
Lopez, Rafael Iñigo T.	QA / Tester, Documentation	• RRLs • Chapters 1, 3-5 • Proofreading / Editing
Pedido, Queen Maegan	QA / Tester, Documentation	• RRLs • Chapters 2-5 • Proofreading / Editing
Santos, Samantha Lui A.	Leader, Developer / Programmer	• Chapters 1-3 • System Architecture, Implementation, and Design